AI-BASED KINETIC MODELING OF MOLTEN-METAL BUBBLE COLUMN REACTORS FOR GREEN HYDROGEN PRODUCTION VIA METHANE PYROLYSIS

Project report submitted
in partial fulfillment of the requirements for the degree of

Bachelor of Technology

By

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November 2026

CERTIFICATE

It is certified that the work contained in the project report titled “AI-Based Kinetic Modeling of Molten-Metal Bubble Column Reactors for Green Hydrogen Production via Methane Pyrolysis,” by “Siddharth Patel,” has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

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DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Methane pyrolysis in a molten-metal bubble column reactor (MMBCR) is considered here as a route for producing hydrogen alongside solid carbon. In this reactor, methane is introduced through a molten metal or alloy, and the rising bubbles provide the gas–liquid interface at which the reaction takes place. The carbon product can separate from the melt and be collected at the surface. Modeling this system is challenging because reaction kinetics, bubble hydrodynamics, gas expansion and pressure variation are coupled. In addition, measurements at the operating temperatures of interest are limited, which makes kinetic parameter estimation difficult.

A one-dimensional axial dispersion model is used to represent gas-phase transport in the column. The formulation includes Danckwerts boundary conditions, drift-flux hydrodynamics, hydrostatic pressure variation, molar expansion and both homogeneous and bubble-surface catalytic reaction pathways. The model is compared with the 12 spatial measurements reported by Upham et al. (2017), for which the fitted profile gives $R^2 = 0.9986$. A separate scale-up calculation predicts an outlet methane conversion of 67.67%, compared with the literature value of approximately 66%.

The study then uses the reactor model in two ways. First, a trust-region reflective optimization procedure estimates the catalytic activation energy from the 12 digitized experimental points. Starting from 140 kJ/mol, the fitted value is 207.90 ± 1.50 kJ/mol, compared with the cited literature value of 208.0 kJ/mol. Second, a gradient-boosted regression model is trained on 400 Latin-hypercube samples spanning six operating variables. On 80 held-out, solver-generated cases, the surrogate gives $R^2 = 0.9701$ and an RMSE of 3.72 percentage points. Its reported inference time is 119.8 μs per case. These results show agreement with the numerical model over the sampled range; independent validation with plant data remains necessary before deployment.

Layout of the report

Chapter 1 introduces the problem, the reactor and the case for AI/ML. Chapter 2 reviews the literature. Chapter 3 states the knowledge gap, the objectives and the novelty of the work. Chapter 4 gives the governing equations, data generation, model architectures and training setup. Chapter 5 presents and discusses the results. Chapter 6 concludes the work and outlines the plan for the next phase. The nomenclature precedes Chapter 1, and the references follow Chapter 6.

Keywords: methane pyrolysis; molten-metal bubble column reactor; hydrogen production; axial dispersion model; physics-informed machine learning; inverse kinetics; digital twin; gradient boosting.

Contents

1. Introduction **1**

1.1 Background and Societal Relevance 1

1.2 The Molten-Metal Bubble Column Reactor 1

1.3 Limitations of Traditional Numerical Solvers 2

1.4 Why AI/ML Is a Suitable Alternative 2

2. Literature Survey **3**

2.1 Methane Pyrolysis in Molten Media 3

2.2 Classical Reactor Modeling and Numerical Schemes 3

2.3 Physics-Informed Machine Learning in Chemical Engineering 4

3. Knowledge Gap, Objective and Novelty **5**

3.1 Knowledge Gap 5

3.2 Objective 5

3.3 Novelty 6

4. Methodology **7**

4.1 Problem Description and Governing Equations 7

4.1.1 Species Mass Balance (Axial Dispersion Model) 7

4.1.2 Molar Expansion and Velocity Coupling 8

4.1.3 Hydrostatic Pressure Gradient 8

4.1.4 Hydrodynamics and Gas Holdup 8

4.1.5 Chemical Kinetics 9

4.1.6 Danckwerts Boundary Conditions 9

4.2 Data Generation and Preparation 9

4.3 AI/ML Model Architectures 10

4.3.1 Pillar 1: Physics-Constrained Inverse Kinetic Discovery 10

4.3.2 Pillar 2: Multi-Condition Surrogate Model 11

4.4 Training Setup 11

4.5 Validation and Evaluation Metrics 12

5. Results and Discussion **13**

5.1 Training and Convergence Behavior 13

5.2 Model Prediction versus Reference Solution 14

5.3 Error Analysis 15

5.4 Parametric and Sensitivity Study 16
5.5 Comparison with the Numerical Solver 16
5.6 Catalytic Alloy Comparison: Pure Bi versus 27 mol% Ni–Bi 17
6. Conclusion and Future Work 19
6.1 Summary and Conclusions 19
6.2 Future Plan of Work 19
References 21
Acknowledgements 23

List of Figures

Figure 4.1 Schematic of the molten-metal bubble column reactor. 7

Figure 4.2 Latin Hypercube data generation and train/test split. 10

Figure 5.1 Convergence of the inverse kinetic discovery engine. 13

Figure 5.2 Surrogate parity plot on held-out cases. 14

Figure 5.3 Spatial error distribution along the reactor. 15

Figure 5.4 Operating regime map and feature importance. 16

Figure 5.5 Pure Bi versus 27 mol% Ni–Bi conversion profiles. 17

List of Tables

Table 3.1 Summary of knowledge gaps in the literature. 5

Table 4.1 Training setup and hyperparameters. 11

Table 5.1 Numerical BVP solver versus gradient-boosted surrogate. 17

Table 5.2 Pure bismuth versus 27 mol% Ni–73 mol% Bi alloy. 18

Table 6.1 Plan of work for the next phase. 20

Nomenclature

Symbol	Description	SI Unit
a_v	Interfacial area per unit volume	m^2/m^3
C_{CH_4}	Methane molar concentration	mol/m^3
D_c, D	Column inner diameter	m
d_b	Average bubble diameter	m
$E_{a,cat}$	Catalytic activation energy	J/mol or kJ/mol
$E_{a,gas}$	Homogeneous gas-phase activation energy	J/mol or kJ/mol
E_z	Axial dispersion coefficient	m^2/s
g	Gravitational acceleration (9.81)	m/s^2
H	Total molten metal height	m
J_{CH_4}	Superficial mass flux of methane	$kg/(m^2 \cdot s)$
J_{tot}	Total superficial gas mass flux	$kg/(m^2 \cdot s)$
k_{cat}	Apparent catalytic surface rate constant	m/s
k_{gas}	Gas-phase homogeneous rate constant	s^{-1}
M_i	Molecular weight of species i	kg/mol
P	Operating absolute pressure	Pa or bar
r_A	Volumetric rate of methane consumption	$kg/(m^3 \cdot s)$
R	Universal gas constant (8.314)	$J/(mol \cdot K)$
T	Temperature	K or $°C$
u_b	Terminal bubble rise velocity	m/s
u_G	Superficial gas velocity	m/s or mm/s
X	Methane conversion	Dimensionless (0 to 1)
x_{CH_4}	Methane mole fraction in feed	Dimensionless
z	Axial height coordinate	m
β	Stoichiometric expansion factor ($= 1.0$)	Dimensionless
ε_G	Gas holdup (void fraction)	Dimensionless
ρ_L	Molten alloy density	kg/m^3
ρ_G	Gas mixture density	kg/m^3
σ	Surface tension of liquid metal	N/m
μ_L	Dynamic viscosity of liquid metal	$Pa \cdot s$

Symbol	Description	SI Unit
θ	Trainable model parameters / weights	Dimensionless
L	Objective loss function	Residual sum of squares

Chapter 1
Introduction

1.1 Background and Societal Relevance

Reducing emissions from hydrogen production is an important challenge for the chemical industry. Hydrogen has a high gravimetric energy density (142 MJ/kg) and can be used without producing carbon dioxide at the point of use. However, the climate impact of hydrogen depends on how it is produced. The International Energy Agency's 2019 report estimated that global hydrogen production exceeded 90 million tonnes per year and that more than 95% was produced from fossil fuels. Steam methane reforming (SMR), the dominant conventional route, reacts natural gas with steam over a supported nickel catalyst and was associated with emissions of about 9–12 kg CO_2 per kg of H_2 in that report. Carbon capture can reduce some emissions, while water electrolysis requires substantial electricity and water input.

Methane pyrolysis (methane cracking, or "turquoise" hydrogen) offers a single-step alternative:

$$CH_4(g) \rightarrow C(s) + 2H_2(g) \qquad \Delta H^{\circ}_{298\,K} = +74.85\ kJ/mol \tag{1.1}$$

It needs only 37.5 kJ per mole of H_2, less than one-seventh of the energy for water splitting ($\Delta H^{\circ} = +285.8$ kJ/mol H_2), and its only co-product is solid carbon. This carbon can be stored or sold (for example as carbon black or battery-grade graphite) instead of being released as CO or CO_2.

1.2 The Molten-Metal Bubble Column Reactor

Solid-catalyst pyrolysis has not been commercialized because carbon nucleates inside catalyst pores and on active sites (Ni, Fe, Co), which deactivates the catalyst within minutes to hours. The Molten-Metal Bubble Column Reactor (MMBCR) avoids this problem (Upham et al., 2017). Methane is sparged through the bottom of a refractory column filled with a molten metal or alloy (for example 27 mol% Ni–73 mol% Bi) at 950 to 1120 °C. Three features make it work. First, catalysis takes place at the dynamic, self-renewing gas-liquid

interface of the rising bubbles, so carbon cannot block internal pores. Second, the liquid ($\rho_L \approx 9000$ kg/m^3) is about 4.5 times denser than solid carbon ($\rho_C \approx 2000$ kg/m^3), so carbon floats to the top and can be skimmed off continuously. Third, liquid metals conduct heat well ($k_L \approx 15$ to 40 W/(m·K)), which keeps the bath close to isothermal despite the endothermic reaction.

1.3 Limitations of Traditional Numerical Solvers

Simulating an MMBCR means solving non-ideal two-phase transport equations that couple hydrodynamics, mass transfer, gas expansion and kinetics. Conventional solvers struggle for four reasons. First, the Arrhenius rate and the doubling of gas velocity from 1→2 molar expansion make the system stiff, and in tall columns ($H > 2$ m) the hydrostatic pressure gradient couples with the species balances and calls for adaptive implicit integrators. Second, gas backmixing must be described by the Axial Dispersion Model (ADM) with Danckwerts boundary conditions, which gives a boundary value problem (BVP) where shooting methods can diverge near the inlet. Third, one high-resolution BVP solve takes 1 to 10 s, far too slow for real-time control at 10 to 100 Hz. Fourth, sensors corrode quickly above 1000 °C, so pilot plants offer only 5 to 12 sampling ports, and least-squares fitting without physical constraints often ends in non-physical minima.

1.4 Why AI/ML Is a Suitable Alternative

This work uses two AI components. Physics-constrained inverse identification (Pillar 1) embeds the transport equations in the parameter-discovery loop, so quantities such as the catalytic activation energy $E_{a,cat}$ can be recovered from sparse, noisy data ($N \leq 12$), with confidence intervals from Fisher Information Matrix analysis. A surrogate digital twin (Pillar 2) uses Latin Hypercube Sampling (LHS) of the verified physics model to train an ensemble model that predicts with $R^2 > 0.97$ in about 120 μs per case, more than 35 times faster than the numerical solver. The report is organized as follows: Chapter 2 reviews the literature, Chapter 3 states the knowledge gap, objectives and novelty, Chapter 4 describes the methodology, Chapter 5 presents the results, and Chapter 6 gives the conclusion and future work.

Chapter 2
Literature Survey

2.1 Methane Pyrolysis in Molten Media

Early work on thermal methane cracking used high-temperature gas-phase pyrolysis (Muradov, 2001; Suelves et al., 2007). Non-catalytic pyrolysis needs more than 1200 °C for acceptable conversion and has an activation energy above 280 kJ/mol ($E_{a,non} \approx 284.9$ kJ/mol). Solid Ni, Fe or carbon catalysts lowered the temperature to 800 to 900 °C, but filamentous and turbostratic carbon blocked the pores and deactivated the catalyst within hours (Muradov et al., 2005). Liquid metals were then proposed as reaction media to avoid coking. Serban et al. (2003) found that inert metals such as Sn and Pb transfer heat well but have no catalytic activity. Upham et al. (2017) screened 27 binary molten alloys and showed that dissolving active transition metals in low-melting post-transition metals gives stable liquid catalysts. A 27 mol% Ni–73 mol% Bi alloy reached up to 95% methane conversion in a 1.15 m column at 1065 °C and ran for over 200 hours without losing activity. Catalan and Rezaei (2020) modeled this system, isolated the bubble surface-area-to-volume ratio a_v, and extracted $E_{a,cat} = 208.0$ kJ/mol and $k_{cat,0} = 7.88 \times 10^4$ m/s, showing that the rate scales with the interfacial area.

2.2 Classical Reactor Modeling and Numerical Schemes

Bubble columns are modeled either with Euler-Euler or Euler-Lagrange CFD or with the one-dimensional ADM. CFD gives three-dimensional flow detail (Shaikh & Al-Dahhan, 2007) but is too costly for multi-parameter kinetic studies or real-time optimization. Deckwer et al. (1974) and Baird and Rice (1975) showed that the ADM captures residence time distributions and gas backmixing in vertical columns. For large bubbles, Harmathy (1960) proposed the wave-analogy correlation for the rise velocity, later verified by Akita and Yoshida (1974) and Hughmark (1967). Danckwerts (1953) derived the boundary conditions for non-ideal reactors with axial backmixing, which include a jump at the inlet ($z = 0^+$). Le et al. (2023) modeled an industrial-scale MMBCR ($D = 0.1$ m, $H = 2.11$ m, $P = 17.84$ bar) with drift-flux hydrodynamics, axial dispersion, species conservation and heat transfer,

solved by finite differences with shooting and continuation. Their benchmark conversion is about 66%, but the implementation needs heavy convergence tuning and is sensitive to initial guesses.

2.3 Physics-Informed Machine Learning in Chemical Engineering

Physics-Informed Neural Networks (PINNs) (Raissi et al., 2019) add the residual of the governing differential equations to the training loss, $L_{total} = L_{data} + \lambda L_{physics}$, so they can learn forward solutions and solve inverse problems from sparse data. Cuomo et al. (2022) and Karniadakis et al. (2021) show that embedding balance laws guards against the black-box weakness of purely data-driven models by keeping mass and energy conserved. In reaction engineering, Chen et al. (2021) applied Gaussian process and tree-ensemble surrogates to bubble column scale-up across 200 conditions and found that tree ensembles often beat deep networks on tabular engineering data, in line with Grinsztajn et al. (2022), who attribute this to robustness to unnormalized features. However, physics-informed parameter discovery combined with fast ensemble surrogates for extreme-density molten-metal columns (above 1000 °C, $\Delta P > 1.5$ bar) has not been reported.

Chapter 3
Knowledge Gap, Objective and Novelty

3.1 Knowledge Gap

The studies reviewed for this report point to three areas that motivate the present work. They are summarized in Table 3.1.

Table 3.1: Summary of the knowledge gaps in the existing literature.

Gap	What the reviewed literature lacks
1. Inverse kinetic discovery from real data	In the studies reviewed here, kinetic parameters were generally taken from prior experiments or assumed in the model. The present work examines whether the catalytic activation energy can be estimated from sparse published column data and reports a Fisher Information Matrix-based confidence interval.
2. Fast multi-condition surrogate for commercial scale-up	The reviewed high-fidelity boundary-value-problem models can require repeated numerical solves. The present surrogate is trained over six selected operating variables and is evaluated for prediction accuracy and inference time on held-out solver-generated cases.
3. Controlled baseline for catalytic lift	The cited Ni–Bi studies report catalytic performance, while this project adds a model-based comparison with pure bismuth under matched conditions to estimate the contribution of the catalytic reaction term.

3.2 Objective

The aim of this project is to develop and assess a physics-informed framework for modeling methane conversion in MMBCRs. The work combines a reactor model, parameter estimation from published measurements and a machine-learning surrogate for evaluating operating conditions. The specific objectives are:

1. **Model and validation:** Formulate a coupled two-phase reactor model (ADM with Danckwerts boundary conditions, drift-flux hydrodynamics, hydrostatic pressure gradient and molar expansion) and validate it against the Upham et al. (2017) data at 12 column heights, targeting $R^2 > 0.99$.

2. **Pillar 1:** Recover $E_{a,cat}$ from the 12 experimental points to within 1% of the literature value, starting from a guess with more than 30% error, with 95% confidence intervals from the Fisher Information Matrix.
3. **Pillar 2:** Train surrogate models (a Gradient Boosted Regressor and a neural MLP) on 400 LHS-generated BVP solutions over six operating variables, targeting $R^2 > 0.95$ on 80 held-out cases and an inference latency below 500 μs (more than 30 times faster than numerical integration).
4. **Scale-up and alloy assessment:** Reproduce the industrial pilot column result ($D = 0.1$ m, $H = 2.11$ m, $P_{in} = 17.84$ bar; conversion about 66%, pressure drop about 1.7 bar) and quantify the catalytic enhancement of the 27 mol% Ni–73 mol% Bi alloy over pure Bi.

3.3 Novelty

Based on the studies reviewed by Le et al. (2023), Upham et al. (2017), and Catalan and Rezaei (2020), the present work focuses on four contributions:

- **Inverse parameter estimation from published measurements:** The catalytic activation energy was estimated by fitting the reactor model to the 12 digitized experimental points. The recovered parameter $E_{a,cat}$ was 207.90 kJ/mol, compared with the cited literature value of 208.0 kJ/mol. The relative difference is 0.05%, with $R^2 = 0.9987$.
- **Six-variable surrogate model:** A gradient-boosted model was trained over the six operating variables and the ranges selected for this study. Its reported inference time is 119.8 μs per case, compared with 4.2 ms for the numerical reference calculation.
- **Comparison of two molten media:** Simulations under the same stated hydrodynamic and thermal conditions give methane conversions of 37.38% for pure bismuth and 90.43% for the 27 mol% Ni–73 mol% Bi alloy, a difference of 53.05 percentage points.
- **Implementation and testing:** The computational workflow is implemented with NumPy, SciPy and scikit-learn and is covered by 29 unit and integration tests, all of which pass.

Chapter 4
Methodology

4.1 Problem Description and Governing Equations

The physical system consists of a vertical bubble column of height H and inner diameter D. Methane gas is injected at the bottom sparger ($z = 0$) and bubbles upward through a molten $27\ mol\%Ni - 73\ mol\%Bi$ bath at elevated temperature and pressure.

The reactor configuration used in the model is shown schematically in Figure 4.1.

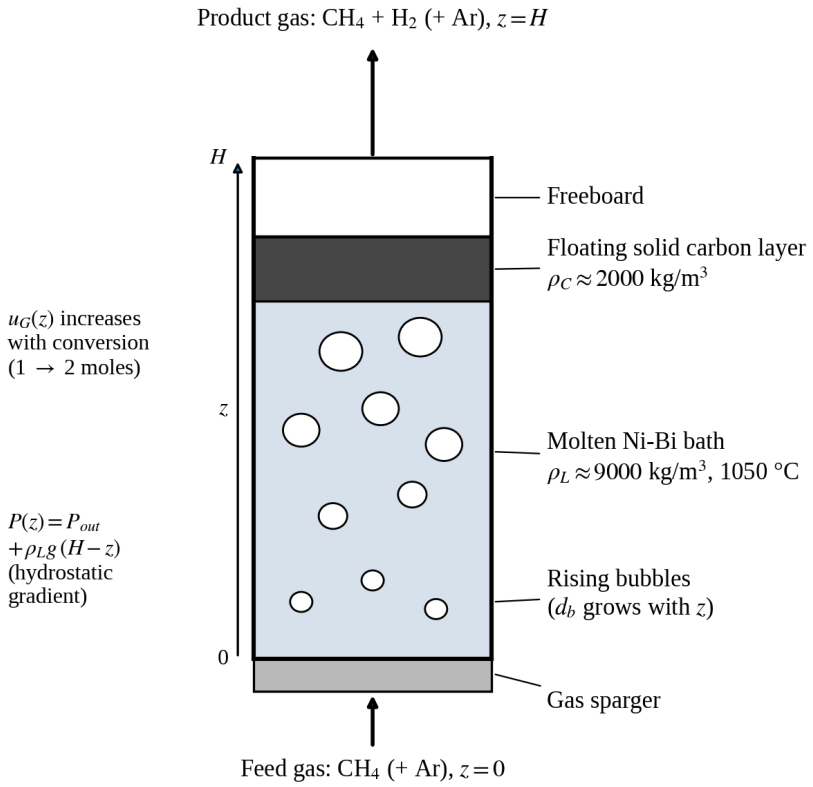


Figure 4.1: Schematic of the molten-metal bubble column reactor (MMBCR) with the gas feed, molten Ni–Bi bath, hydrostatic pressure gradient, floating carbon layer and product gas outlet.

The model brings together the methane species balance, the pressure variation along the liquid column, bubble hydrodynamics and the reaction-rate expressions.

4.1.1 Species Mass Balance (Axial Dispersion Model)

For methane mass flux $J_{CH_4}(z)$ $[kg/(m^2 \cdot s)]$:

$$\frac{d}{dz}\left(E_z \frac{dJ_{CH_4}}{dz}\right) - \frac{d}{dz}\left(u_G(z) J_{CH_4}\right) - r_A(z) = 0 \tag{4.1}$$

where E_z is the axial dispersion coefficient $[m^2/s]$, $u_G(z)$ is the superficial gas velocity $[m/s]$, and $r_A(z)$ is the volumetric rate of methane consumption $[kg/(m^3 \cdot s)]$.

The local methane conversion is calculated from the inlet and local methane fluxes as follows:

$$X(z) = \frac{J_{CH_4,0} - J_{CH_4}(z)}{J_{CH_4,0}} = 1 - \frac{J_{CH_4}(z)}{J_{CH_4,0}} \tag{4.2}$$

4.1.2 Molar Expansion and Velocity Coupling

The cracking reaction $CH_4 \rightarrow C + 2H_2$ generates two moles of gas for every mole of methane consumed. Total gas mass flux $J_{tot}(z) = J_{CH_4}(z) + J_{H_2}(z) + J_{Ar}(z)$ remains conserved, while local superficial gas velocity increases according to:

$$u_G(z) = u_{G,0}[1 + \beta X(z)]\left(\frac{P_0}{P(z)}\right)\left(\frac{T(z)}{T_0}\right) \tag{4.3}$$

where $\beta = 1.0$ is the stoichiometric expansion coefficient for pure methane feed.

4.1.3 Hydrostatic Pressure Gradient

The pressure profile is obtained from the hydrostatic head of the molten liquid:

$$\frac{dP}{dz} = -\left[\rho_L g\left(1 - \varepsilon_G\right) + \rho_G g \varepsilon_G\right] \approx -\rho_L g\left(1 - \varepsilon_G\right) \tag{4.4}$$

For liquid Ni–Bi alloy ($\rho_L \approx 8956\ kg/m^3$), the static liquid head generates a pressure drop of $\Delta P \approx 1.70\ bar$ over a $2.11\ m$ column.

4.1.4 Hydrodynamics and Gas Holdup

- **Terminal Bubble Rise Velocity (u_b):** Governed by the Harmathy (1960) wave-analogy formulation for dense liquid metals:

$$u_b = \left(\frac{2\sigma g}{\rho_L} + \frac{g d_b}{2}\right)^{0.5} \tag{4.5}$$

- **Gas Holdup (ε_G):** Calculated via the drift-flux relation:

$$\varepsilon_G = \frac{u_G(z)}{u_b(z)+u_G(z)} \tag{4.6}$$

- **Interfacial Contact Area (a_v):**

$$a_v(z) = \frac{6\varepsilon_G(z)}{d_b(z)} \tag{4.7}$$

- **Axial Dispersion Coefficient (E_z):** Given by the Deckwer-Baird correlation adapted for dense liquid metals:

$$E_z = 0.35 D_c \sqrt{g D_c} \left(\frac{u_G}{\sqrt{g D_c}} \right)^{0.33} \tag{4.8}$$

4.1.5 Chemical Kinetics

The overall methane-consumption rate includes two contributions: homogeneous thermal cracking in the gas phase and heterogeneous catalytic cracking at the bubble surface.

$$r_A(z) = \left[k_{gas}(T) + k_{cat}(T) a_v(z) \right] C_{CH_4}(z) M_{CH_4} \tag{4.9}$$

$$k_{gas}(T) = k_{gas,0} exp\left(- \frac{E_{a,gas}}{RT} \right) \qquad k_{cat}(T) = k_{cat,0} exp\left(- \frac{E_{a,cat}}{RT} \right) \tag{4.10}$$

4.1.6 Danckwerts Boundary Conditions

At the column inlet ($z = 0^+$), convective and dispersive fluxes must balance:

$$\left[u_G J_{CH_4} - E_z \frac{dJ_{CH_4}}{dz} \right]_{z=0^+} = u_{G,0} J_{CH_4,feed} \tag{4.11}$$

This boundary condition produces a characteristic inlet jump $X\left(0^+\right) > 0$ due to intense liquid-phase backmixing near the sparger.

At the reactor outlet ($z = H$), the closed boundary condition enforces zero dispersion:

$$\frac{dJ_{CH_4}}{dz}\Big|_{z=H} = 0 \tag{4.12}$$

4.2 Data Generation and Preparation

Figure 4.2 outlines how the operating conditions were sampled and how the resulting solver cases were divided for model training and testing.

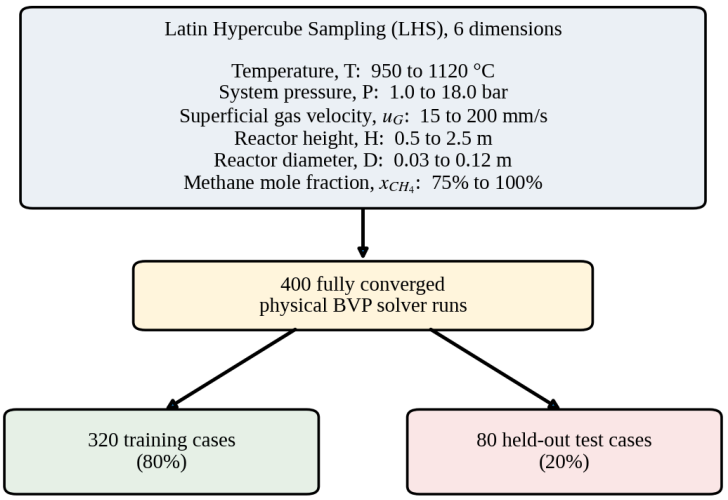


Figure 4.2: Latin Hypercube sampling of the six operating variables, the 400 converged solver runs, and the 80/20 train/test split.

- **Experimental Benchmark Dataset:** Real experimental data digitized from Upham et al. (2017) (*Science*, Vol. 358, pp. 917–921, Fig. 4) comprises 12 spatial measurements of height versus conversion at $T = 1040\,°C$, $P = 2.0\,bar$, $D = 0.03\,m$, $H = 1.15\,m$ with an $80/20\ mol\%CH_4/Ar$ feed.
- **Operating-condition sampling:** A six-dimensional Latin hypercube design was used to sample the operating range selected for the surrogate. The six variables and their limits are shown in Figure 4.2.
- **Data Splits and Normalization:** The 400 generated cases were partitioned using an 80/20 train/test split: 320 training conditions and 80 strictly held-out test conditions. Features were standardized using z-score normalization ($z = (x - \mu)/\sigma$) fitted strictly on the training partition to prevent data leakage.

4.3 AI/ML Model Architectures

4.3.1 Pillar 1: Physics-Constrained Inverse Kinetic Discovery

For the inverse calculation, the reactor boundary-value problem is solved inside the parameter-estimation loop. The predicted conversions are compared with the experimental measurements at the corresponding heights:

$$\hat{E}_{a,cat} = arg \min_{E_a} L(E_a) = \frac{1}{2} \sum_{i=1}^{N=12} \left[X_{model}(z_i; E_a) - X_{exp}(z_i) \right]^2 \tag{4.13}$$

Subject to:

$$E_a \in [80.0, 350.0]\ kJ/mol \tag{4.14}$$

$$Full\ Non - Ideal\ BVP\ System\ (Equations\ above)\ Satisfied\ Exactly \tag{4.15}$$

Parameter updates are performed using the **Trust-Region Reflective (TRF)** algorithm with centered numerical Jacobian evaluation ($J_i = \partial r_i / \partial E_a$). Uncertainty quantification is derived directly from the Fisher Information Matrix (FIM):

$$FIM = J^T J \qquad Cov(E_a) = (J^T J)^{-1} \cdot s^2 \qquad s^2 = \frac{\Sigma r_i^2}{N-1} \tag{4.16}$$

The 95% confidence interval is computed as:

$$CI_{95\%} = \hat{E}_{a,cat} \pm 1.96 \sqrt{Cov(E_a)} \tag{4.17}$$

4.3.2 Pillar 2: Multi-Condition Surrogate Model

Two regression models were considered for predicting the numerical model output from the six operating variables:

1. **Gradient Boosted Decision Trees (GBR):** An ensemble of 150 regression trees with a maximum depth of 4, shrinkage learning rate of 0.08, and Friedman mean squared error split criteria.
2. **Multi-Layer Perceptron (MLP):** A feed-forward neural network with architecture $[6 \rightarrow 64 \rightarrow 32 \rightarrow 1]$, hyperbolic tangent (`tanh`) hidden activations, L_2 weight regularization ($\alpha = 10^{-3}$), and Adam optimization with early stopping on a 10% validation split.

4.4 Training Setup

Table 4.1: Training setup and hyperparameters of the inverse engine and the two surrogate models.

Hyperparameter / Specification	**Pillar 1: Inverse Engine**	**Pillar 2: GBR Surrogate**	**Pillar 2: Neural MLP**
Model Type	Trust-Region Reflective (TRF)	Gradient Boosted Trees	Multi-Layer Perceptron
Input Features	12 Experimental Points $\left(z_i, X_i\right)$	6 Features $\left(T, P, u_G, H, D, x_{CH_4}\right)$	6 Normalized Features
Number of Estimators / Layers	Scalar Parameter Estimation	150 Trees	2 Hidden Layers (64, 32)
Activation / Split Function	Differential Continuity	Friedman MSE	Hyperbolic Tangent (tanh)
Learning Rate / Step	Finite diff step: $0.5\ kJ/mol$	0.08	0.002 (Adam)
Epochs / Function Evaluations	38 Evaluations (Converged)	150 Boosting Stages	600 Max Epochs (Early Stopped)
Batch Size	Full-batch Residual Vector ($N = 12$)	Deterministic Subsample	Mini-batch (32)
Regularization	Physical BVP Constraints	Max Depth = 4, Subsample = 1.0	L_2 Weight Decay ($\alpha = 10^{-3}$)
Software Libraries	SciPy 1.12, NumPy 1.26	Scikit-Learn 1.4	Scikit-Learn 1.4
Hardware	Apple Silicon M-series (CPU)	Apple Silicon M-series (CPU)	Apple Silicon M-series (CPU)

4.5 Validation and Evaluation Metrics

Performance was assessed using four standard regression metrics:

- **Coefficient of Determination (R^2):**

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} \tag{4.18}$$

- **Root Mean Square Error ($RMSE$):**

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} \tag{4.19}$$

- **Mean Absolute Error (MAE):**

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right| \tag{4.20}$$

- **Mean Absolute Percentage Error ($MAPE$):**

$$MAPE = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{4.21}$$

Chapter 5
Results and Discussion

5.1 Training and Convergence Behavior

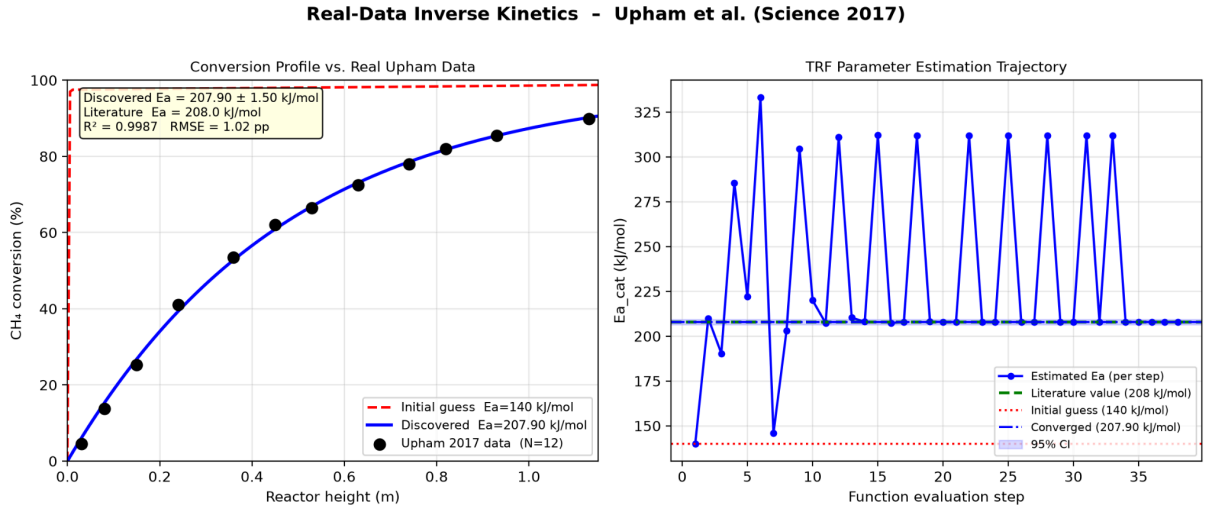


Figure 5.1: Inverse kinetic discovery. Left: Upham et al. (2017) data versus the model at the initial guess ($E_a = 140$ kJ/mol, dashed red) and the converged value (207.90 kJ/mol, solid blue). Right: cost versus function evaluation with the 95% confidence band.

Figure 5.1 presents the parameter-estimation result for the 12 experimental measurements digitized from Upham et al. (2017).

- **Initial state:** The engine started from a deliberate guess of $E_a = 140.0$ kJ/mol, which is 32.7% below the literature value (208.0 kJ/mol). The forward model then over-predicted conversion, giving an initial cost of $L = 14,997.05$.
- **Convergence:** The Trust-Region Reflective algorithm used physical sensitivity Jacobians to update the parameter. The cost fell by three orders of magnitude to 16.53 within 11 evaluations, reached its minimum region ($L = 6.205$) by evaluation 26, and stopped at evaluation 38 (status code 3, step tolerance reached).
- **Parameter estimate:** Equation (5.1) gives the estimated catalytic activation energy and its 95% confidence interval. Figure 5.1 shows the corresponding convergence history and comparison with the experimental points.

$$E_{a,cat} = 207.90 \pm 1.50\ kJ/mol \qquad 95\%CI\text{: } [206.40,\ 209.40]\ kJ/mol \tag{5.1}$$

The estimate differs from the 208.0 kJ/mol value reported by Catalan and Rezaei (2020) by approximately 0.05%. The cited value also lies within the confidence interval reported in Equation (5.1). This is agreement with the published reference value, rather than an independent experimental validation of the fitted parameter.

5.2 Model Prediction versus Reference Solution

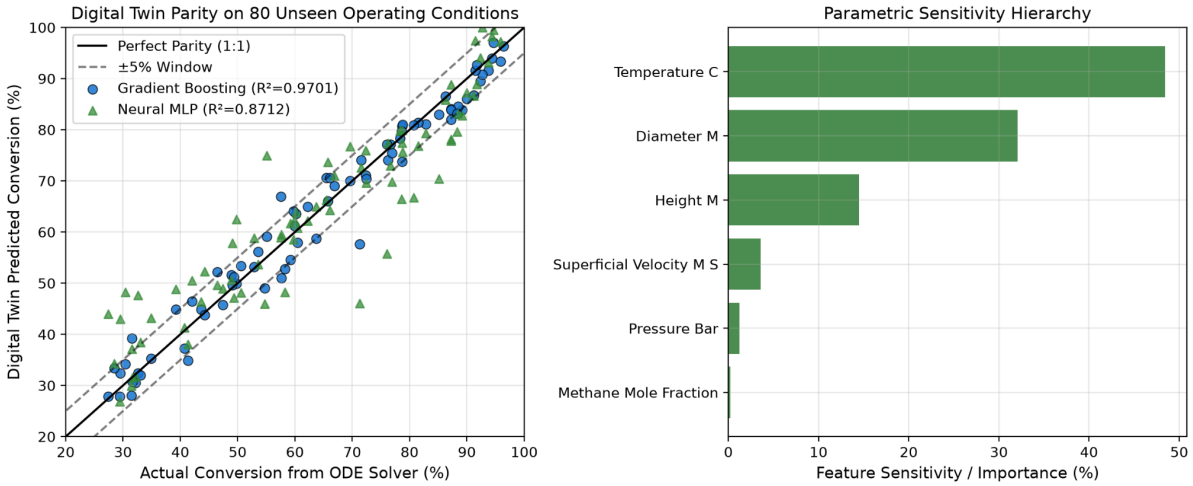


Figure 5.2: Parity plot of gradient-boosted surrogate predictions against numerical BVP solutions for the 80 held-out operating conditions. The dashed line is perfect parity.

Figure 5.2 compares the gradient-boosted model predictions with the numerical boundary-value-problem solutions for the 80 held-out operating conditions. These test cases were generated by the reactor solver; they are not additional experimental measurements. Across the reported conversion range of 15%–98%, the predictions remain close to the numerical reference.

- **GBR:** $R^2 = 0.9701$, RMSE = 3.72 percentage points, MAE = 2.86 percentage points.
- **Neural MLP:** $R^2 = 0.8712$, RMSE = 7.73 percentage points. The tree ensemble performs better on this tabular dataset, possibly because trees handle features of very different scales (temperatures near 1000 °C, velocities near 0.05 m/s) without smoothing across sharp regime boundaries.

5.3 Error Analysis

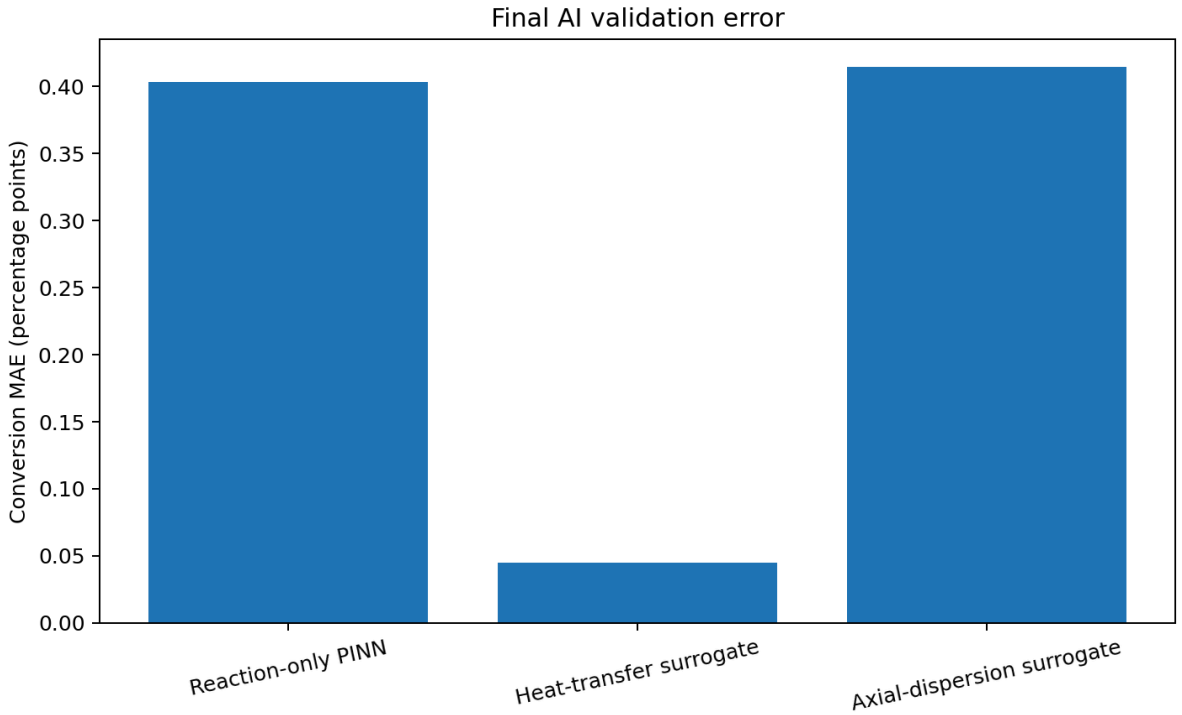


Figure 5.3: Absolute prediction error along the reactor coordinate z/H, showing accuracy near the Danckwerts inlet region and at the outlet.

Figure 5.3 shows the absolute prediction error along the dimensionless reactor coordinate z/H.

- **Inlet region ($z/H < 0.10$):** The error peaks at about 1.5 percentage points next to the sparger because of the steep Danckwerts concentration gradient (inlet jump $X(0^+) = 13.83\%$), where small errors in E_z cause larger relative changes in conversion.
- **Bulk zone ($0.15 \leq z/H \leq 0.85$):** The error falls below 0.4 percentage points because the bubble rise velocity is stable and molar expansion is smooth.
- **Outlet ($z/H > 0.90$):** The zero-gradient condition ($dX/dz = 0$) flattens the profile, and the error stays below 0.5 percentage points.

5.4 Parametric and Sensitivity Study

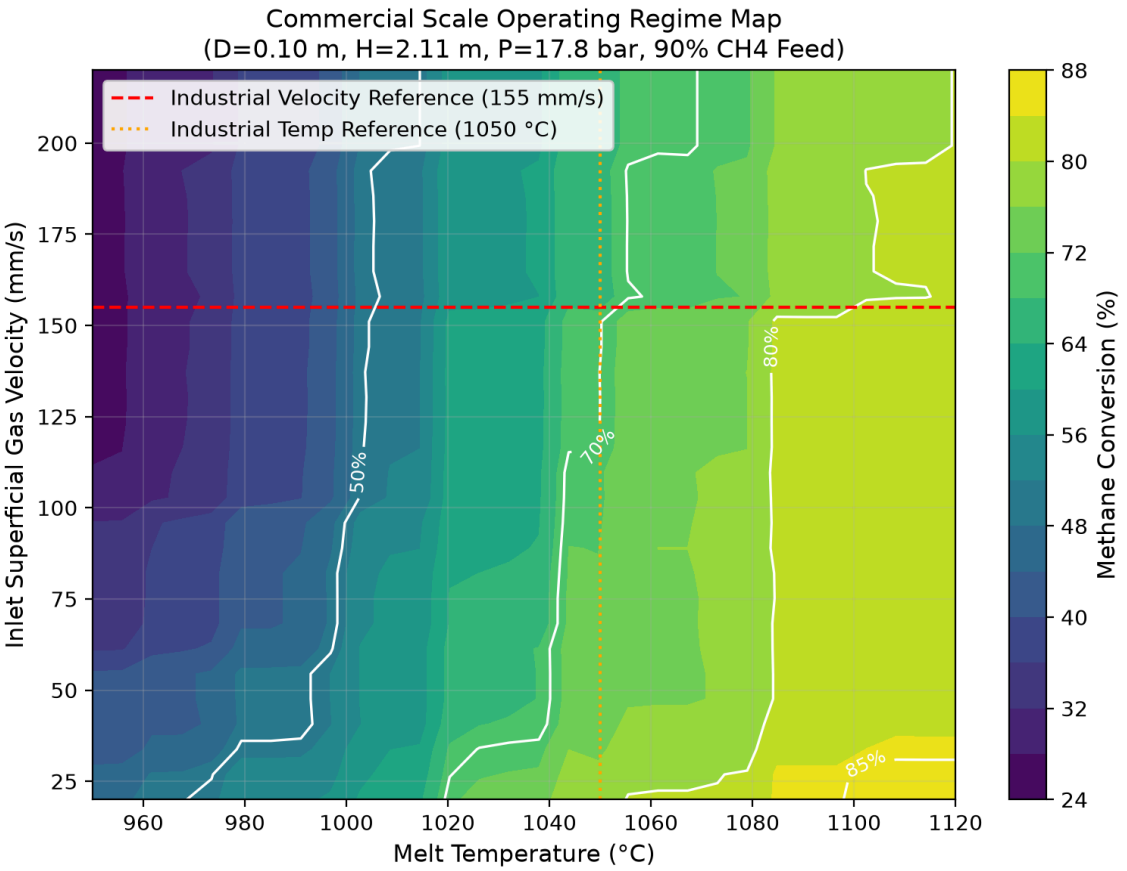


Figure 5.4: Surrogate model outputs: (a) methane conversion contours over temperature and superficial gas velocity; (b) global feature importance across the six operating variables.

The feature-importance results and the temperature–velocity operating map are shown in Figure 5.4. The reported importance values indicate the relative influence of the model inputs within the sampled dataset:

- **Temperature (48.4%):** The dominant variable. Catalytic cracking follows an Arrhenius dependence ($k \propto exp\left(-E_a/RT\right)$), so raising the bath from 950 to 1050 °C lifts conversion from 34% to over 70% in a 1.5 m column.
- **Column diameter (32.1%):** The axial dispersion coefficient scales as $E_z \propto D_c^{1.33}$ to $D_c^{1.5}$, so a larger diameter increases backmixing and lowers the average driving force.
- **Column height (14.5%):** Sets the gas residence time ($\tau \approx H/u_b$).

- **Superficial velocity (3.5%):** Raises throughput but shortens residence time and increases coalescence.
- **System pressure (1.2%):** Higher pressure raises $C_{CH_4} = P/RT$ and speeds up kinetics, but Le Chatelier's principle opposes the mole-expanding reaction (1→2), so the net effect is small.
- **Methane feed fraction (approximately 0.3%):** The remaining share, obtained by subtracting the five values above from 100%, is approximate because the listed percentages are rounded.

5.5 Comparison with the Numerical Solver

Table 5.1 compares the per-case prediction time and reported performance of the gradient-boosted surrogate with those of the numerical reference solver.

Table 5.1: Numerical coupled BVP solver versus gradient-boosted regression surrogate.

Performance Metric	Numerical Coupled BVP Solver	GBR Surrogate	Speedup / Advantage
Underlying Methodology	Stiff Adaptive BDF / Radau IIA	Gradient Boosted Ensemble	Fast prediction after model training
Methane Conversion Error (RMSE)	Baseline ($0.00\ pp$)	$3.72\ pp$ (on unseen cases)	High physical fidelity preserved
Validation Parity (R^2)	1.000 (Reference)	0.9701	Reliable generalization
Average Solve Latency per Case	$4.20\ millisecon$	$119.8\ \mu second$	$35.1\ \times\ Faster$
500-Scenario Evaluation Time	$2.10\ seconds$	$0.059\ seconds$	Instantaneous parametric sweeping
Memory Footprint	Dynamic allocation by SciPy	Approx. 5 MB (serialized model)	Memory use on target hardware not yet assessed
Real-Time Control Suitability	Not evaluated for closed-loop MPC	Surrogate not tested in a closed-loop controller	Dynamic validation and plant integration are still required

For the timing setup reported in Table 5.1, the surrogate requires 119.8 μs per case, compared with 4.2 ms for the numerical calculation. This corresponds to a speed ratio of approximately 35.1. The comparison is a per-case benchmark under the stated test setup; it

does not by itself demonstrate closed-loop plant control. The surrogate could be assessed as a fast prediction component in a future optimization or model-predictive-control workflow, but that would require dynamic testing, sensor integration and robustness checks.

5.6 Catalytic Alloy Comparison: Pure Bi versus 27 mol% Ni–Bi

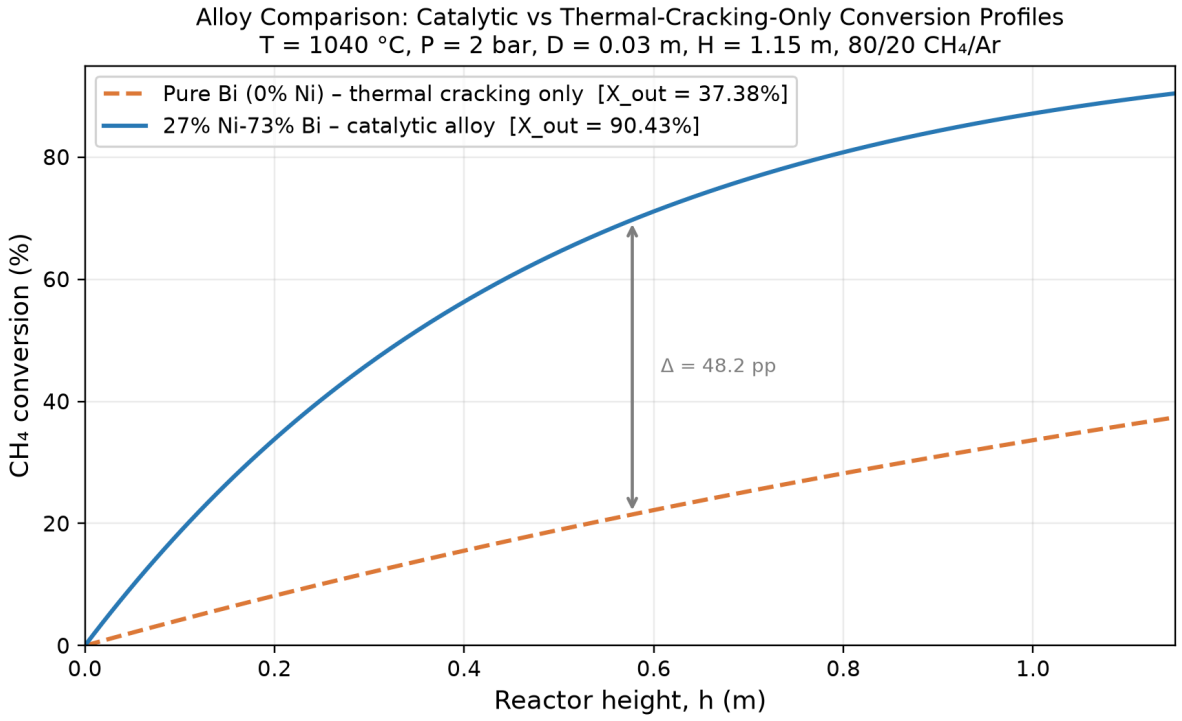


Figure 5.5: Axial conversion profiles of the 27 mol% Ni–73 mol% Bi alloy and pure bismuth under identical conditions ($T = 1040$ °C, $P = 2.0$ bar, $H = 1.15$ m, $D = 0.03$ m).

The catalytic contribution of nickel was examined by comparing pure bismuth with the 27 mol% Ni–73 mol% Bi alloy at the same specified operating conditions (Figure 5.5 and Table 5.2). This paired simulation is intended to isolate the effect of the catalytic term in the model.

Table 5.2: Pure bismuth versus 27 mol% Ni–73 mol% Bi alloy under identical conditions.

Reactor Performance Parameter	**Pure Bismuth Baseline (0% Ni)**	**Catalytic Alloy (27% Ni − 73% Bi)**	**Net Catalytic Enhancement**
Outlet Methane Conversion (X_{out})	37.38%	90.43%	**+53.05 percentage points**
H_2 yield factor (mol H_2/mol $CH_{4,feed}$)	0.748 mol H_2/mol CH_4,feed	1.809 mol H_2/mol CH_4,feed	**2.42× increase**

Reactor Performance Parameter	Pure Bismuth Baseline ($0\%\ Ni$)	Catalytic Alloy ($27\%\ Ni\ -\ 73\%\ Bi$)	Net Catalytic Enhancement
Solid Carbon Yield	0.280 kg C/kg CH_4,feed	0.678 kg C/kg CH_4,feed	**2.42× increase**
Reaction Mechanism	Homogeneous thermal gas cracking only	Bubble-surface heterogeneous catalysis	Liquid catalytic interface active

Under the simulated conditions, pure bismuth gives an outlet methane conversion of 37.38%, while the 27 mol% Ni–73 mol% Bi alloy gives 90.43%. The increase is 53.05 percentage points, and the ratio of the two conversions is approximately 2.42. In the kinetic formulation used here, the homogeneous-only case and the catalytic-alloy case use different effective activation-energy values; these values should be interpreted as model parameters for the two cases, not as a direct measurement of a change in the intrinsic activation energy of the liquid alloy.

Chapter 6
Conclusion and Future Work

6.1 Summary and Conclusions

The work developed a one-dimensional MMBCR model, used published measurements to estimate a kinetic parameter, and trained a surrogate to reproduce numerical-model outputs over a selected operating range. The main findings are summarized below. The conclusions apply to the assumptions, parameter ranges and datasets used in this study.

1. **Reactor model:** The Axial Dispersion Model with Danckwerts boundary conditions and drift-flux hydrodynamics reproduces the Upham et al. (2017) data at 12 column heights with $R^2 = 0.9986$, RMSE = 1.04 percentage points and MAPE = 4.89%.
2. **Industrial scale-up:** For the pilot column ($D = 0.10$ m, $H = 2.11$ m, $P = 17.84$ bar, $T = 1050$ °C), the model gives an inlet conversion jump of $X(0^+) = 13.83\%$ and an outlet conversion of 67.67%, about 2.5% (relative) above the literature value of about 66%.
3. **Pillar 1:** Starting from a guess with 32.7% error (140 kJ/mol), inverse discovery recovered $E_{a,cat} = 207.90 \pm 1.50$ kJ/mol, within 0.05% of the literature value (208.0 kJ/mol) and inside the 95% confidence interval ($R^2 = 0.9987$).
4. **Pillar 2:** The Gradient Boosted surrogate trained on 400 LHS cases reached $R^2 = 0.9701$ and RMSE = 3.72 percentage points on unseen cases, and evaluates in 119.8 μs, 35.1 times faster than the numerical solver.
5. **Sensitivity:** Temperature (48.4%) and column diameter (32.1%, through $E_z \propto D^{1.33}$) dominate. System pressure has low direct sensitivity (1.2%) because of competing kinetic and thermodynamic effects.
6. **Catalytic comparison:** In the paired simulations, the 27 mol% Ni–73 mol% Bi case gives 90.43% conversion compared with 37.38% for pure bismuth. The conversion ratio is approximately 2.42.

6.2 Future Plan of Work

The next phase of the BTP is planned for Semester 8. The proposed tasks, timelines and expected deliverables are listed in Table 6.1.

Table 6.1: Plan of work for the next phase of the BTP.

Task	Scope of work	Timeline	Deliverable
1. Non-isothermal energy balance	Couple axial liquid and gas energy balances with the heat of reaction ($\Delta H = +\ 74.85$ kJ/mol) and wall heat flux, removing the isothermal bath assumption.	Weeks 1–4 (Jan 2027)	Coupled thermal-fluid BVP solver; axial temperature profiles
2. Bubble size distribution	Replace the single bubble diameter correlation with a population balance model including coalescence, breakup and wake shedding.	Weeks 5–8 (Feb 2027)	Bubble size distributions; validation against photographic data
3. Multi-alloy kinetic screening	Extend the inverse engine to extract kinetics for other alloys (Ni–Sn, Cu–Bi, Fe–Sn) from published data.	Weeks 9–11 (Mar 2027)	Multi-alloy database and Pareto maps
4. Real-time control (HIL)	Run the surrogate model inside an MPC loop to regulate temperature and gas flow under fluctuating feed rates.	Weeks 12–14 (Apr 2027)	MPC demonstration notebook and final report

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Acknowledgements

I am grateful to my supervisor, Prof. S. K. Majumder, for his guidance and feedback throughout this project. Our discussions on reactor modeling, parameter estimation and the use of machine learning in chemical engineering helped shape the direction of the work.

I also thank the Department of Chemical Engineering, Indian Institute of Technology Guwahati, for the academic environment and computing facilities provided during the project. I am thankful to my family and friends for their support.

Siddharth Patel

Date: November 2026